MILITARY SPECIFICATION

COMPONENTS TEST JUNCTION ASSEMBLY

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the performance and acceptance inspection requirements for the Components Test Junction Assembly associated with the M-130 Aircraft General Purpose Dispenser.

2. APPLICABLE DOCUMENTS

2.1 The following documents of issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

FSC: 4925

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies Procedures For
MIL-STD-810 - Environmental Test Methods

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND
9317991 - Components Test Junction Assembly
9321404 - Packing and Marking for Box, Exterior for Components Test Junction Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing 9317991 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Circuit resistance. - The maximum circuit resistance shall be 0.2 ohms between the points in Table 1 when tested in accordance with 4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

TABLE 1

FROM	TO	FROM	TO	FROM	TO
P2-A	P1-d	P2-R	P1-X	P2-t	P4-W
P2-A	P6-C	P2-S	P1-Y	P2-t	P5-P
P2-B	P1-p	P2-T	P1-Z	P2-BB	TP-GND
P2-B	P6-M	P2-U	P1-C	P2-BB	P2-9
P2-C	P1-e	P2-V	P1-D	P2-BB	P6-R
P2-D	P1-f	P2-W	P1-E	P2-BB	P5-R
P2-D	P6-F	P2-X	P1-F	P4-G	P5-S
P2-D	P6-P	P2-Y	P1-G	P5-B	P4-a
P2-E	P1-g	P2-Z	P1-a	P6-J	TP-J
P2-E	P6-A	P2-a	P1-S	P6-K	TP-K
P2-F	P1-h	P2-b	P1-T	P6-L	TP-L
P2-F	P6-B	P2-c	P1-U	P6-N	TP-N
P2-G	P1-K	P2-d	P1-V	P6-T	P3-1
P2-G	P6-D	P2-e	P1-W	P6-T	P4-B
P2-H	P1-L	P2-f	P1-A	P6-T	P5-A
P2-H	P1-S	P2-g	P1-B	P6-T	J1-1
P2-J	P1-M	P2-DD	P6-G	P6-U	P4-E
P2-K	P1-N	P2-CC	P3-2	P6-U	J1-5
P2-L	P1-P	P2-CC	P3-6	P3-5	P4-R
P2-M	P1-R	P2-t	P4-A	P3-5	P5-M
P2-N	P1-i	P2-t	P3-8	P7-1	P4-H
P2-p	P1-j	P2-t	P6-H		

MODE SELECT	CHAFF-FLARE COUNT	DCP FLARE COUNT	FROM	TO
Chaff-MAN	Chaff	OFF	P2-y	P6-T
Chaff-MAN	Chaff	OFF	P2-y	J2-Center Conductor
Flare-MAN	Flare	ON	P2-y	J1-5
Flare-MAN	Flare	ON	P2-y	J2-Center Conductor
Flare-MAN	Flare	ON	P2-y	P3-5
Flare-RIPPLE	Flare	ON	P2-y	P4-G
Flare-RIPPLE	Flare	ON	P2-y	P5-S
Chaff-PGRM	Chaff	ON	P2-y	P4-a
Chaff-PGRM	Chaff	ON	P2-y	P5-B

NOTE: TP = TEST POINT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

3.4 Insulation resistance. - The insulation resistance of each isolated pin to all other pins (See Table 2) electrically connected together shall be 20 megohms minimum when tested in accordance with 4.5.2.

TABLE 2

P2 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, f, g, t, y, BB, CC, DD

P6 - J, K, T, L, U, N, H, R

P4 - G, R, a

Place switches in the following position during insulation resistance tests.

Mode Select Switch	Chaff-MAN
Chaff-Flare Count	Chaff
DCP Flare Count	OFF

3.5 Transportation vibration. - The component test junction assembly shall maintain circuit resistance (3.3) and insulation resistance (3.4) after being subjected to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X. The assembly shall be vibrated along three mutually perpendicular axes.

3.6 Wiring and soldering. - Soldering of electrical wires, cables, connectors, components and assemblies shall be performed in a thorough, workmanlike manner in accordance with MIL-STD-1460.

3.7 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Components Test Junction Assembly	9317991	2

4.3.2 Inspections to be performed. - See MIL-A-48078 and paragraphs 4.4.2.1 through 4.4.2.4, testing per paragraph 4.4.3, and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
	Components Test Junction Assembly	1	1	9317991
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects		100%	3.2	4.4.2.4
	Circuit Resistance		100%	3.3	4.4.3.1
	Insulation Resistance		100%	3.4	4.4.3.2
	Transportation Vibration		100%	3.5.1	4.4.3.3
	Examination for Defects		100%	3.2	4.4.2.4
	Circuit Resistance		100%	3.3	4.4.3.1
	Insulation Resistance		100%	3.4	4.4.3.2

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cover, Assembly		9321210
			NEXT HIGHER ASSEMBLY 9317991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Color of connectors, electrical test points		0.40 %	3.2	Visual
102	Soldering improper or inadequate		0.40%	3.6	4.4.3.4/Visual
Minor					
201	Wire marking missing, inadequate or incorrect		0.65%	3.2	Visual
202	Cover marking missing, inadequate or incorrect		0.65%	3.2	Visual
203	Protective finish missing or inadequate		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Box Assembly		9321274
			NEXT HIGHER ASSEMBLY 9317991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Soldering improper or inadequate		0.40%	3.6	4.4.3.4/Visual
Minor					
201	Wire marking missing, inadequate or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Cable Assembly, Special Purpose, Electrical		9321197
			NEXT HIGHER ASSEMBLY
			9317991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Soldering improper or inadequate (prior to assembly)		0.40%	3.2	4.4.3.4/Visual
103	Insulation missing or inadequate		0.40%	3.4	Visual
Minor					
201	Wire marking missing, adequate or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Components Test Junction Assembly				9317991
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Identification plate missing or improperly annotated		0.40%	3.2	Visual
103	Soldering improper or inadequate		0.40%	3.6	4.4.3.4/Visual
104	Switch not aligned with markings		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cover	1 OF 1	9321209
			NEXT HIGHER ASSEMBLY 9321210

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Size and location of holes incorrect		0.40%	3.2	Gage
102	Marking missing or incorrect		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Box		9321215
			NEXT HIGHER ASSEMBLY 9321274

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of holes on mounting bracket		0.40%	3.2	Gage
102	Protective finishing missing or inadequate		0.40%	3.2	Visual
103	Marking missing or incorrect		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63149(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Exterior for Components Test Junction Assembly				9321404
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Barrier bag punctured or inadequately sealed		0.40%	5.0	Visual
Minor					
201	Contents loose		0.65%	5.0	Manual
202	Marking missing, illegible or incorrect		0.65%	5.0	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Circuit resistance. - Major Defect - Each test junction assembly shall be subjected to this test. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. If any unit fails to complywith the applicable requirements, it shall be classified defective and removed from the lot.

4.4.3.2 Insulation resistance. - Major Defect - Each test junction assembly shall be subjected to this test. The test shall be performed as specified in 4.5.2 using equipment in accordance with 4.4.4. If any unit fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.4.3.3 Transportation vibration. - Major Defect - Each test junction assembly submitted for first article testing shall be subjected to this test. This test shall be performed as specified in 4.5.3 using equipment in accordance with 4.4.4. If any unit fails to comply with the applicable requirements, the first article sample shall be rejected.

4.4.3.4 Improper soldering. - Major Defect - Each test junction assembly shall be inspected for improper soldering accordance with 3.6. If any wiring or printed circuit card assembly fails to comply with the requirements of 3.6, the unit shall be classified defective and removed from the lot.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/ Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures

4.5.1 Circuit resistance. - Electrical measurements of the test junction assembly shall be performed in accordance with Method 302 of MIL-STD-202.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

4.5.2 Insulation resistance. - Electrical measurements of the test junction assembly shall be performed in accordance with Method 302, Test Condition B of MIL-STD-202.

4.5.3 Transportation vibration. - Using equipment in accordance with 4.4.4, subject each first article unit to the vibration requirements of MIL-STD-810, Method 514.2, Procedure X to verify conformance to 3.5. Following this test, inspect each test junction assembly for evidence of damage. Any dispenser which fails to comply to the applicable requirements shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing. - The assemblies shall be packed as specified on Dwg. 9321404.

5.2 Marking. - Marking of the box shall be in accordance with Dwg. 9321404.

6. NOTES

6.1 Intended use. - This test junction assembly is used in testing the Aircraft General Purpose Dispenser, M-130.

6.2 Ordering data. - Procurement documents shall specify the following:

a. Title, number and data of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - (See MIL-A-48078) Submit equipment designs as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-Q, Dover,New Jersey 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 4925-A-030

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:25Z
Check the source to verify that this is the current version before use.